

PHYSIOLOGY LABS AT HOME



Mark Kelly
Irvine Valley College

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Prelude from Author

Necessity is the mother of invention

The development of this lab manual is actually an interesting story. I have been fascinated with physiological assessments for a very long time. I have always enjoyed finding out about my own physiological capacities or my current physiological status from relatively simple, and relatively non-invasive tests. I use the word “relatively” because plugging what seems like a gas mask on someone, asking them to wear a cold strap around their chest, then run on treadmill without turning their head too much, and running until they are about to explode IS INVASIVE to many. Understandably so! However, no blood has been taken, no organ tissues have been sampled, and no drugs had to be given.

With the onset of COVID19, many of us had to do many things we never expected. Educational classes and labs previously on ground were forced online. I taught an anatomy class where the students needed to order “organs” and dissection kits from a company. We then performed online dissections, with many choosing to use their backyards to avoid the smell in their houses! Even after COVID19 isolation was finished, many of these labs would be a valuable supplement to an on-ground laboratory course.

The main point of a laboratory exercise is to show the students in a “hands on” manner what they learned in lecture. This is very important not only to the tactile-kinesthetic and visual learner, but to future health care workers who will need to know laboratory or surgical techniques. Demonstrating laboratory activities without being in a lab is difficult at best. Various *virtual laboratory applications* have been used for quite some time previous to COVID19, and many have expanded the applications since then. Still, seeing it happen in reality and real time is more powerful! Seeing is believing! I hope you see your own physiology while doing these exercises/discussions/reports.

Inspiration from Devastation or Desperation

I started my company, *Principle-Centered Health*, because of my fascination with assessments. I also started it because Hurricane Katrina forced me out of New Orleans, and I knew that I could use the devastation as inspiration. I brought a lot of fancy equipment at significant prices, in order to test people in exercise and health related areas. I realized that there were a lot of tests requiring little to no equipment, and by simply using a calculator or a smartphone someone could find out a great deal about their own physiology in a wide variety of systems. I would make many of these easy to perform labs as extra credit for my classes.

Fast forward to the COVID and post-COVID era. With the onset of online laboratories, many people had to purchase the virtual lab applications and use the internet in very innovative ways. Many instructors had to be quite innovative with “home-made” labs. The beauty or inspiration that comes from desperation is that many scientific phenomena are right in front of our eyes on a daily basis! The ability to observe and use science to learn about ourselves or improve our quality of life is what labs should be all about. I hope this lab manual will help you achieve this and show you the many resources you have “at home”. I hope you enjoy it and learn something about yourself in the process.

Dr. Mark Kelly

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